Uniform Plane Wave Introduction

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Introduction to Uniform Plane Waves

- Let field $\vec{E} = \hat{a}_y E_o \cos(\omega t - \beta z)$ –in free space (No free charges) – y-axis
- β – positive real number
- E is a function of time and space
- We have to find $\vec{B}$, $\vec{D}$ and $\vec{H}$ vectors
- We have to show that $\vec{E}$ and $\vec{H}$ constitutes a wave and propagates in z – direction
- $D = \epsilon_o E \Rightarrow D = \hat{a}_y \epsilon_o E_o \cos(\omega t - \beta z)$
- To find B and H, use Maxwell's equation
- Find curl of E

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

- To find Curl (E) → replace H with E in following equation

$$\text{curl } \mathbf{H} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix}$$

- $E = a_y E_o \cos(\omega t - \beta z)$

$$\nabla \times \vec{E} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_o(\cos\omega t - \beta z) & 0 \end{vmatrix} = \frac{\partial}{\partial z} E_o \cos(\omega t - \beta z)(-\hat{a}_x)$$

$$= -\hat{a}_x E_o \beta \sin(\omega t - \beta z)$$

$$= -\frac{\partial \vec{B}}{\partial t}$$

$$\Rightarrow \vec{B} = -\hat{a}_x \frac{E_o \beta}{\omega} \cos(\omega t - \beta z)$$

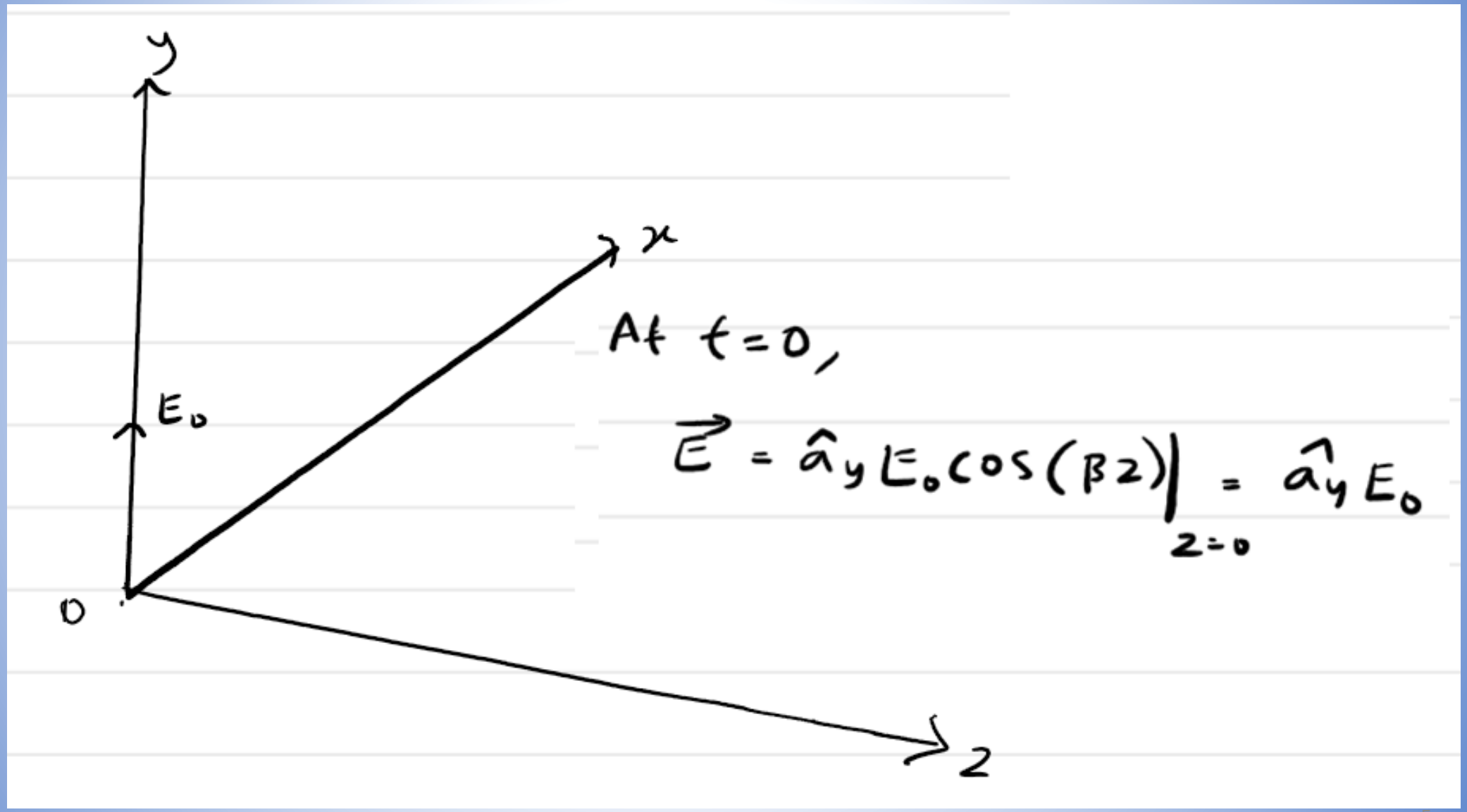
$$\Rightarrow \boxed{\vec{B} = -\hat{a}_x \frac{E_0 \beta}{\omega} \cos(\omega t - \beta z)}$$

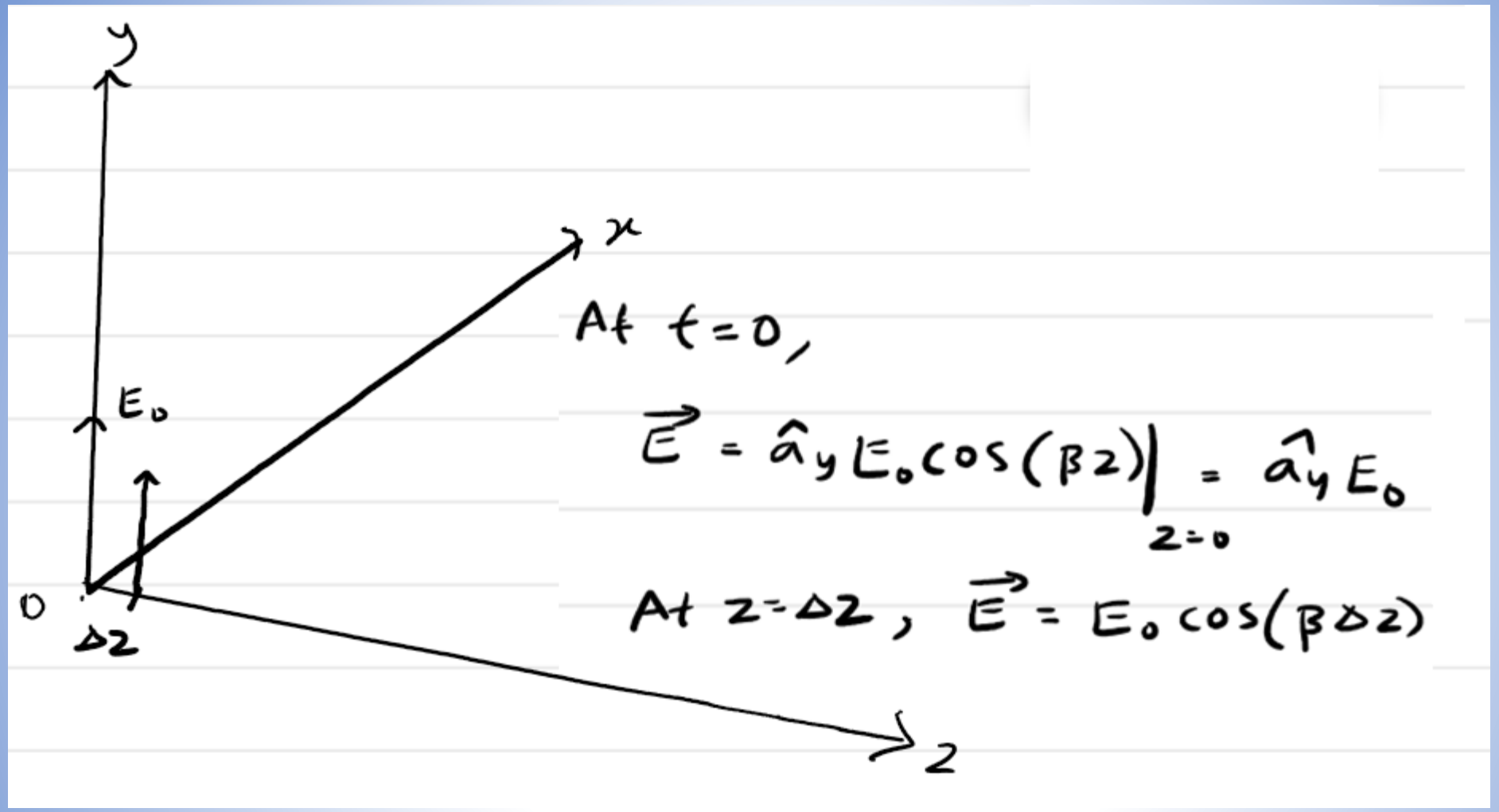
$$\text{WKT } \vec{B} = \mu_0 H \Rightarrow \boxed{H = -\hat{a}_x \frac{E_0 \beta}{\omega \mu_0} \cos(\omega t - \beta z)}$$

$$\boxed{\vec{E} = \hat{a}_y E_o \cos(\omega t - \beta z)}$$

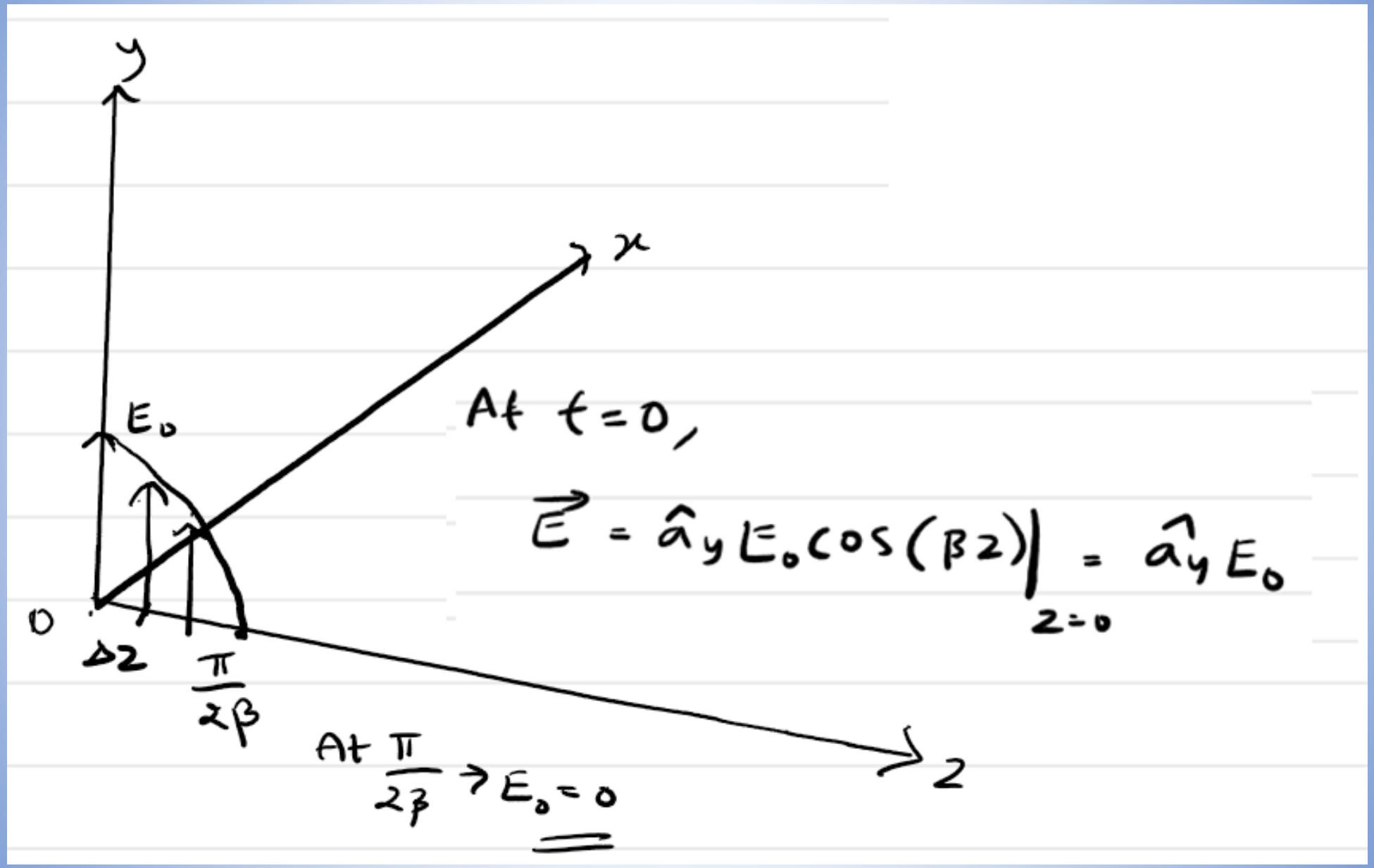
- Now, we have to show that these form a wave propagating in Z direction
- Sketch these fields for 2 close instances of time
- Time t = 0 on Z-axis

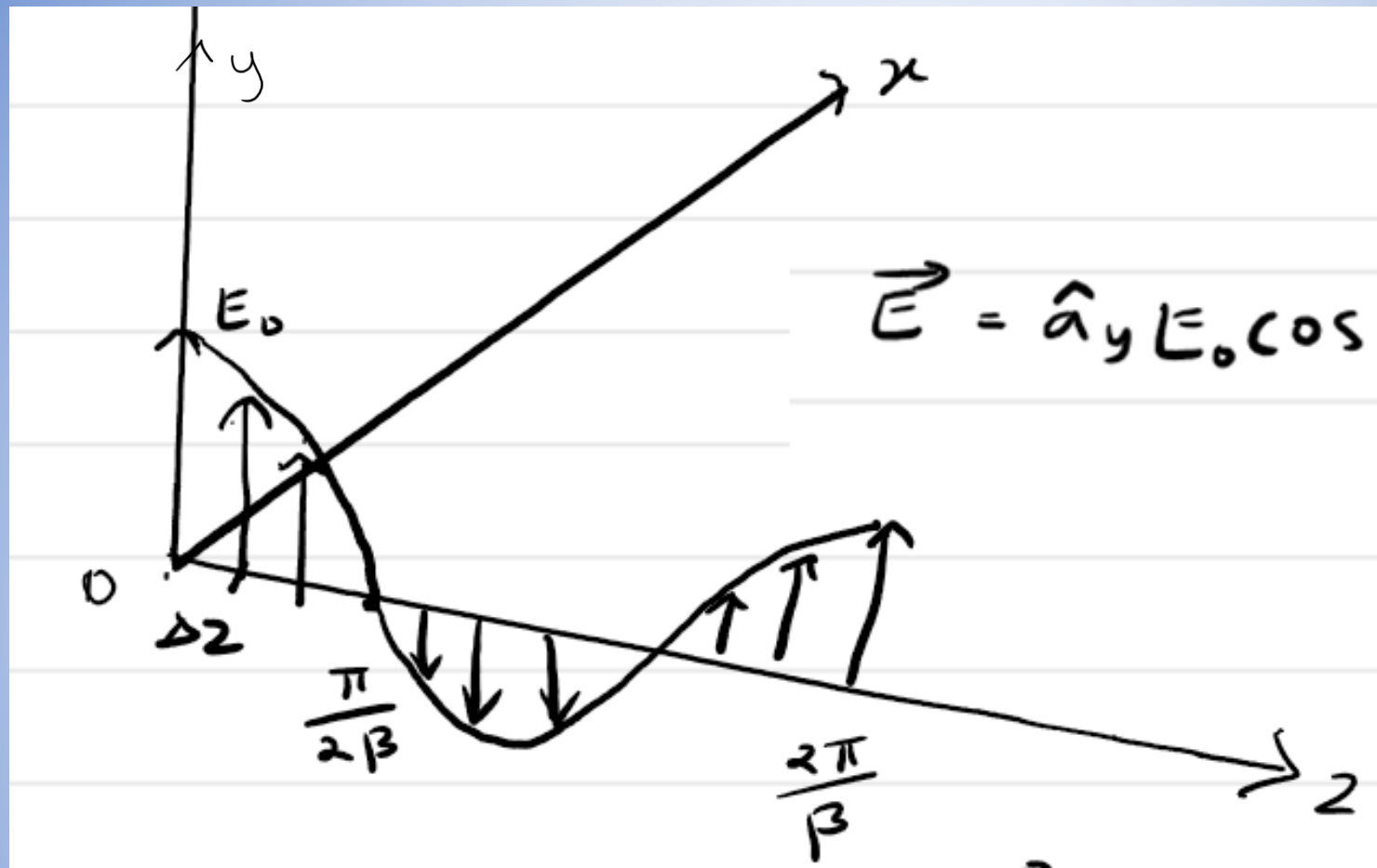
$$\vec{E} = \hat{a}_y E_o \cos(\omega t - \beta z)$$





Smaller than E_0


y
x
E_0
0
Δz
$\frac{\pi}{2\beta}$
z
At $t=0$,
$\vec{E} = \hat{a}_y E_0 \cos(\beta z)\Big|_{z=0} = \hat{a}_y E_0$
At $\frac{\pi}{2\beta} \rightarrow E_0 = 0$



$$\vec{E} = \hat{a}_y E_0 \cos(\beta z)\Big|_{z=0} = \hat{a}_y E_0$$

At $z = \frac{2\pi}{\beta} \Rightarrow \vec{E} = E_0$

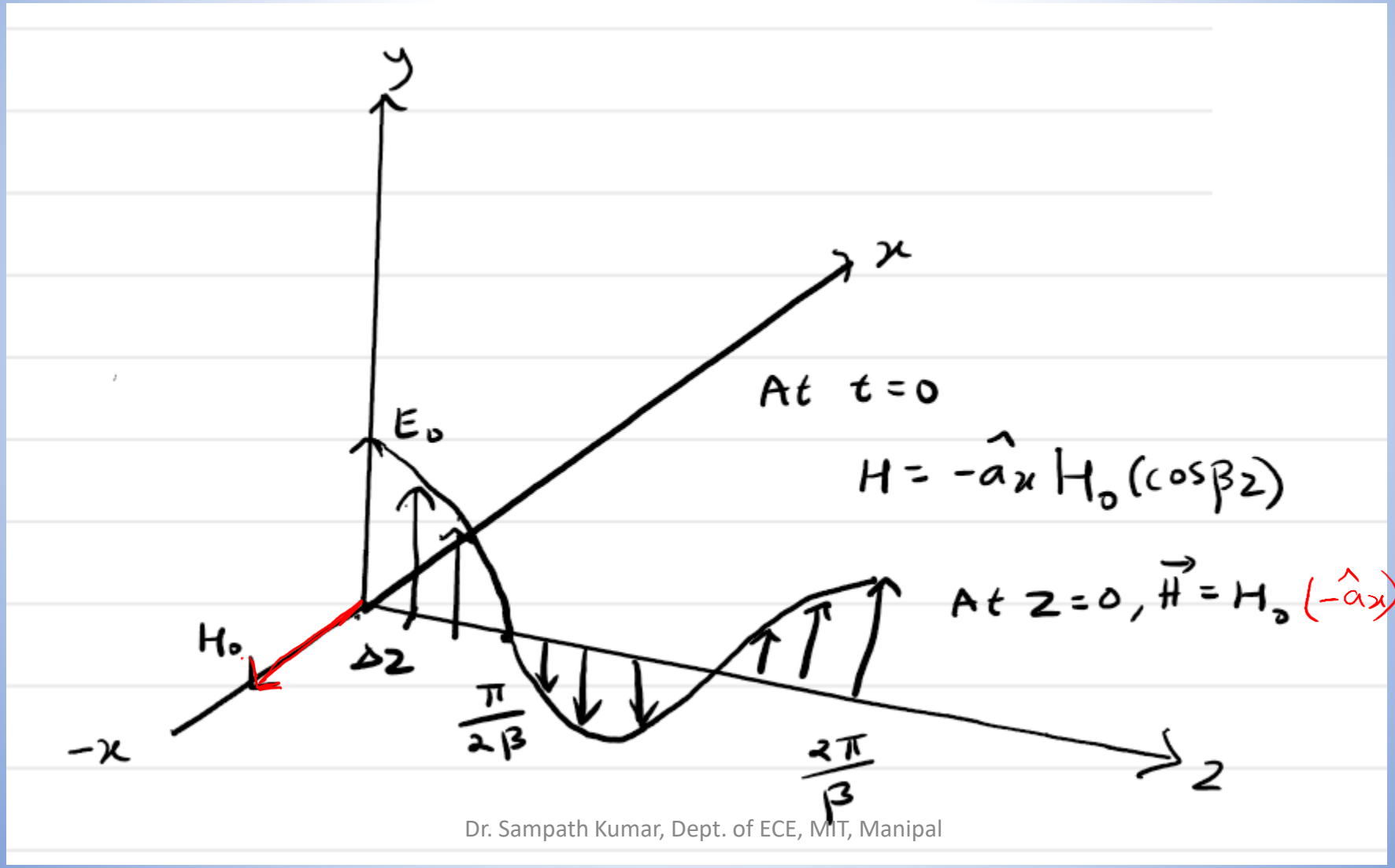
$$\lambda = \frac{2\pi}{\beta}$$

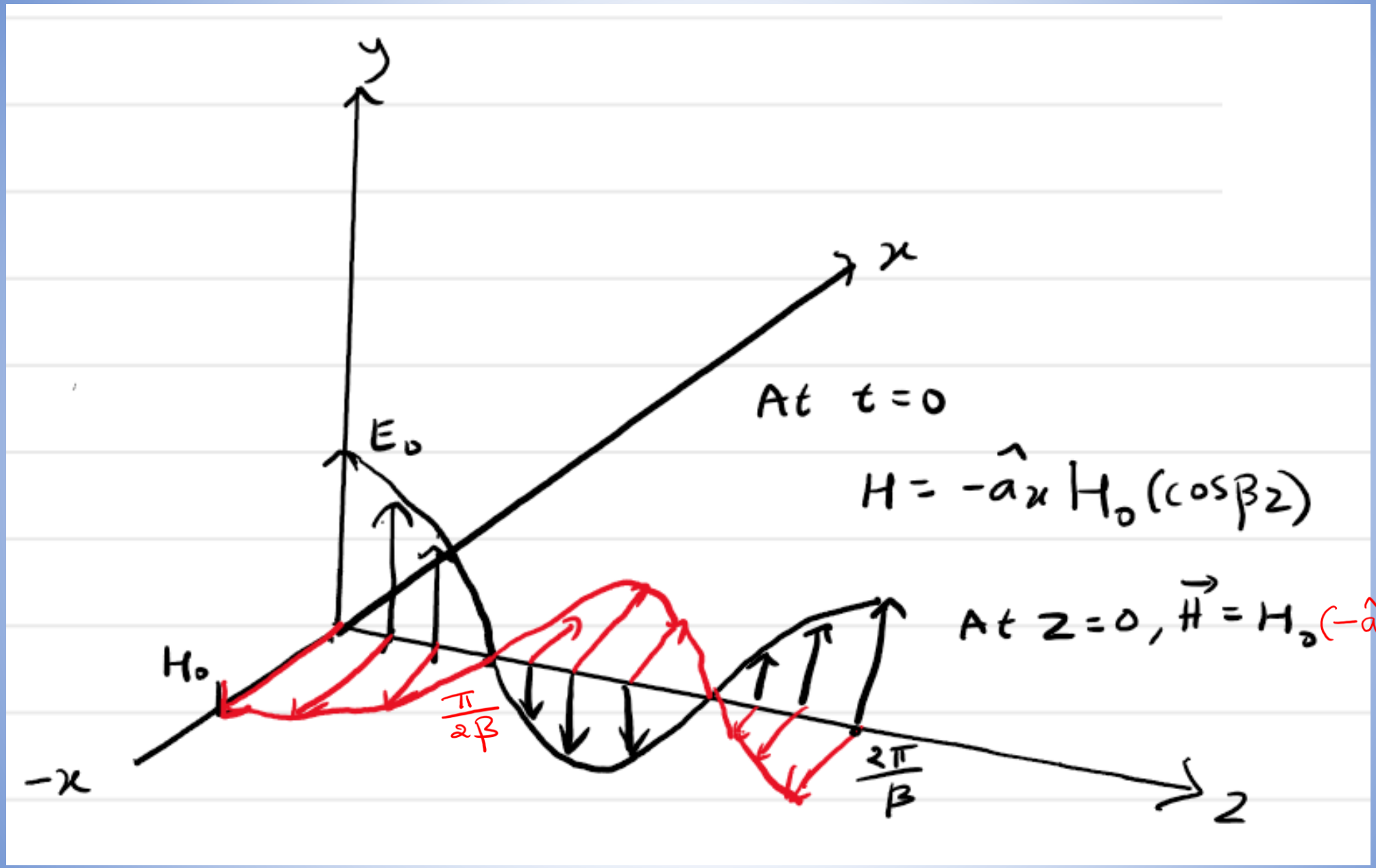
wavelength

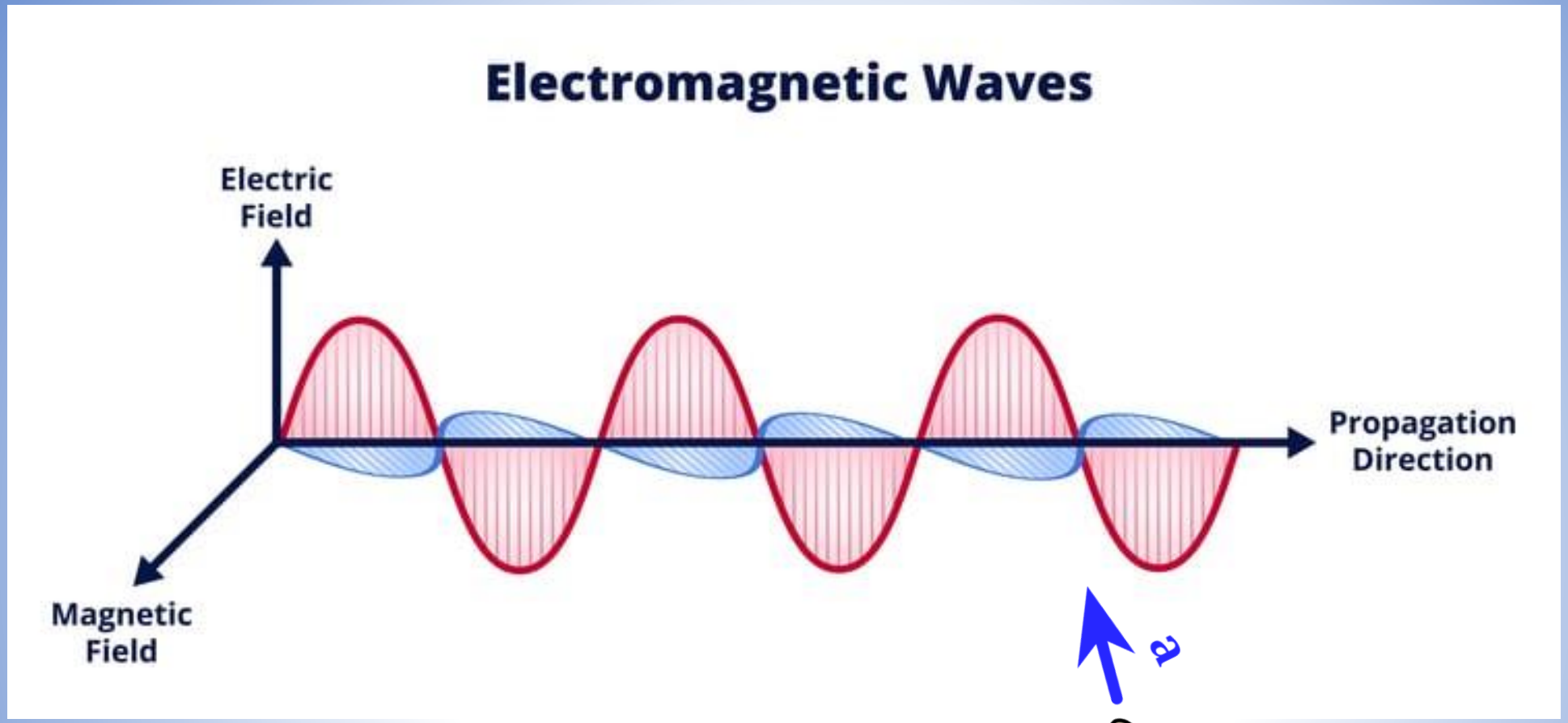
$$\beta = \frac{2\pi}{\lambda}$$

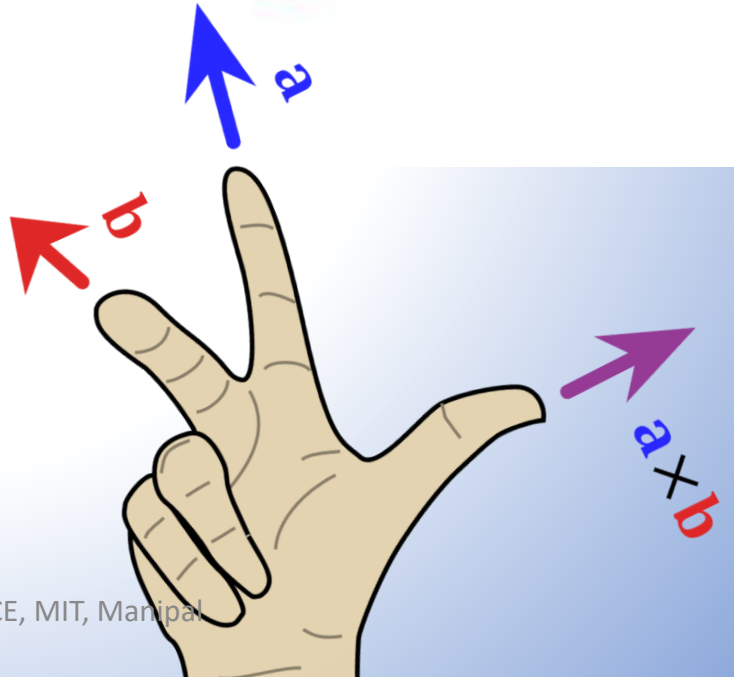
$$H = -\hat{a}_x \frac{E_0 \beta}{\omega \mu_0} \cos(\omega t - \beta z)$$

Let $\frac{E_0 \beta}{\omega \mu_0} = H_0 \Rightarrow \vec{H} = -\hat{a}_x H_0 \cos(\omega t - \beta z)$

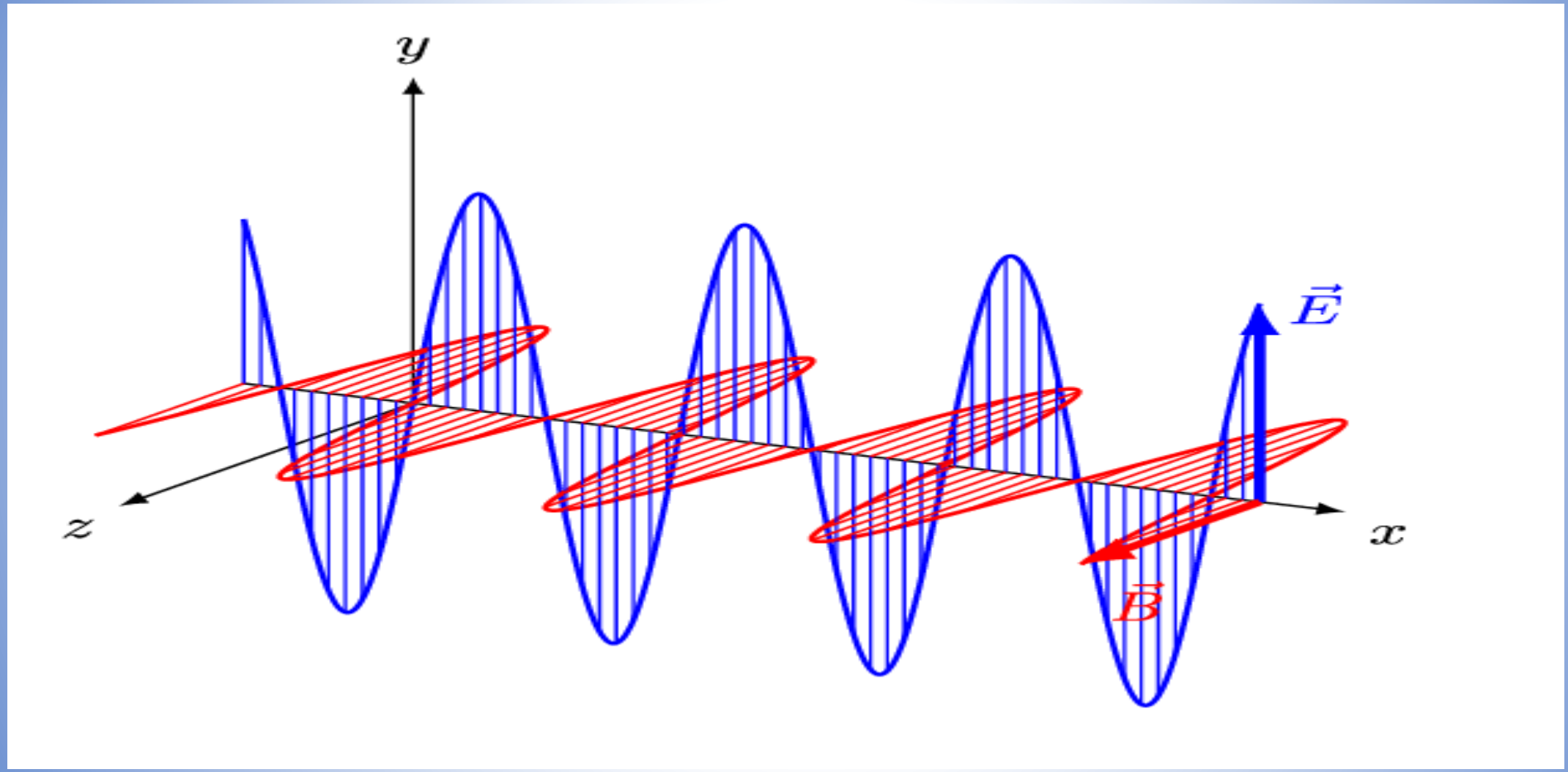



y
x
E_0
H_0
$\frac{\pi}{2\beta}$
$\frac{2\pi}{\beta}$
-x
z
At t = 0
$H = -\hat{a}_x H_0 (\cos \beta z)$
At z = 0, $\vec{H} = H_0(-\hat{a}_x)$





- E x H gives direction of propagation


y
z
x
$\vec{E}$
$\vec{B}$


Thank
you